



2.305 GHz Circular Patch Antenna for the PocketQube Parent Board

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Inspirefly, PocketQube Parent Antenna Board – 2.305 GHz S-band*

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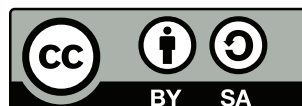
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PocketQube Parent Antenna Board – Inspirefly

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0.9	C. G.	Import patch to KiCAD, DXF flow	2026/03/01
0.5	C. G.	CircularPatch / Hexagonal-Patch HFSS	2025/12/04
0.2	C. G.	Square aperture calculations	2025/11/20



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CHAPTER 1

Introduction

The PocketQube form factor constrains the entire spacecraft to a $50 \times 50 \times 50$ mm cube per unit (1P), with mass below 250 g. For the Inspirefly parent spacecraft, which relays telemetry and payload data from deployed child PocketQubes, the available exterior area for antennas is limited to one face of the structure. A low-profile, directive antenna that can be integrated directly into the printed circuit board (PCB) is therefore preferable to deployable monopoles or large ground-plane-dependent structures.

This document records the design of a 2.305 GHz centered circular microstrip patch antenna for the PocketQube parent antenna board. The frequency was selected to provide an S-band downlink/uplink channel separate from UHF telemetry, with sufficient bandwidth for higher data rates and with regulatory compatibility in the 2.3–2.4 GHz region. The antenna is realized as a probe-fed circular patch on a multilayer PCB, with two orthogonal feed points intended for circular polarization through an external quadrature power divider.

1.1 Why a Circular Patch

Rectangular and circular patches share the same operating principle – a resonant cavity formed by the metal patch, the grounded dielectric substrate, and the fringing fields at the edges – but differ in mode structure, polarization control, and fabrication tolerance. The circular geometry was chosen for three reasons.

First, the dominant TM_{11} mode of a circular patch has a field distribution with azimuthal symmetry, which simplifies generation of circular polarization by feeding at two points separated by 90° in space and driven in quadrature. For a rectangular patch, circular polarization requires either a nearly square patch with truncated corners or a dual-feed with unequal slot perturbations; both are more sensitive to etching tolerances at 2.3 GHz than the symmetric circular case.

Second, for a given resonant frequency the circular patch occupies less area than the circumscribed square patch and has no sharp corners where current crowding increases loss. On a 50 mm board, a radius near 19–20 mm (see Chapter 2) leaves a usable ground-plane ring for connectors and keep-out, while a $\lambda_g/2$ square patch with the same substrate would require a larger footprint once fringing corrections are included.

Third, the project history included trials of a coplanar monopole, a rectangular patch with a slot perimeter equal to λ_g , and a hexagonal patch (commits 99f9ccd, 5b98260, fad1ff9). The hexagonal geometry was evaluated in ANSYS HFSS as a low-area alternative but showed stronger excitation of higher-order modes and less predictable axial ratio. The circular patch provided the most repeatable match to the cavity model and was retained for the final board revision (CircularPatchV3.aedt).

1.2 Objectives

The design objectives for the parent antenna board are:

1. Resonant frequency $f_0 = 2.305$ GHz, with $S_{11} < -10$ dB across at least 20 MHz.
2. Realized gain greater than 5 dBi at boresight (zenith) and axial ratio below 3 dB for the circularly polarized configuration.
3. Input impedance near $50\ \Omega$ at each coax feed, compatible with the Molex 732511350 U.FL-compatible connector and the external Mini-Circuits-style divider APS-02-WBM-LP-DF-CC.
4. Full integration into the KiCAD PCB stack without post-assembly tuning stubs, using the same fabrication stack available for the rest of the spacecraft electronics (FR-408HR multilayer).
5. Reproducibility: all geometry originates from the ANSYS HFSS parametric model and is exported via DXF to the KiCAD footprint `Inspire-fly:CircularPatchAntenna`, with version control in git.

1.3 Document Structure

Chapter 2 develops the underlying theory – cavity model, effective radius, and feed-point impedance – and describes the three tool chains used: MATLAB Antenna Toolbox for rapid particle-swarm exploration, ANSYS HFSS (via AEDT files in `ansys/`) for full-wave finite-element verification, and KiCAD 10.0 for layout (`KiCAD/antennaBoard/`). Chapter 3 reports the numerical results from each stage, including the limits of the early rectangular-patch estimate and the current best MATLAB optimization log (`matlab/best.txt`). Chapter 4 interprets those results and records design decisions visible in git history, such as the move from a single MMCX feed to two orthogonal coax feeds and the substrate change from Rogers RO4350B to FR-408HR. Chapter 5 summarizes status and remaining verification steps before fabrication.

1.4 How to Read This Document

This document is written for a mixed audience. **Mechanical and software engineers** will find the key takeaway in each section's first paragraph; the deeper RF detail follows for the antenna team. **Business development** can read the boxed summaries and Chapter 5 for cost, schedule, and risk. A few terms recur:

- **PocketQube (1P)** – a 50 mm cube satellite standard; think of it as a very small, low-cost spacecraft where every square millimeter counts.
- **S-band (2.3 GHz)** – a radio band that carries more data than UHF but still works with low-cost ground stations.
- **Patch antenna** – a flat metal disc (the patch) printed on a circuit board over a ground plane; it radiates perpendicular to the board with no moving parts.
- **Probe-fed / quadrature / circular polarization** – two small feed points on the disc, driven 90° out of phase, make the radio wave spin. A spinning wave is more robust when the satellite tumbles.
- S_{11} / **return loss** / **gain** / **axial ratio** – measures of match (< -10 dB is good), signal strength (> 5 dBi target), and spin purity (< 3 dB target).

Business takeaway (Introduction). The antenna is a single-sided, PCB-integrated part with no deployment mechanism. That removes a common failure mode, keeps assembly time low, and uses the same low-cost FR-408HR board stack as the rest of the avionics. Frequency 2.305 GHz was chosen to add a higher-rate S-band link without redesigning the UHF system, preserving schedule and ground-station investment.

CHAPTER 2

Methodology

Reader guide. This chapter explains how the antenna was sized and built. Each subsection opens with a plain-language summary for systems, mechanical, software, and business readers; the technical detail and equations follow for the RF team. Boxed notes flag cost, supply, and schedule impact.

This chapter states the analytical model used to size the antenna, the material and stackup choices, and the three-stage numerical flow that links MATLAB exploration, ANSYS HFSS full-wave simulation, and KiCAD layout. All equations are given in final form; no hedging is applied to the procedure.

2.1 Circular Patch Theory

What it is: Treat the patch and ground as a shallow resonant cavity (like a drumhead). Energy at the right frequency bounces efficiently and radiates. The rest of this section quantifies that resonance and the correction for fields that spill beyond the metal edge.

2.1.1 Cavity Model and Resonant Frequency

A circular microstrip patch of physical radius a on a grounded substrate of thickness h and relative permittivity ϵ_r is modeled as a cylindrical cavity with electric walls at the patch and ground plane and a magnetic wall at the periphery [1, 2]. The resonant frequencies of the TM_{nm} modes are

$$f_{nm} = \frac{\chi_{nm} c}{2\pi a_e \sqrt{\epsilon_r}}, \quad (2.1)$$

where $c = 2.99792458 \times 10^8 \text{ m s}^{-1}$, χ_{nm} is the m -th zero of the derivative $J'_n(x)$ of the Bessel function of the first kind, and a_e is the effective radius that accounts for fringing. For the dominant mode TM_{11} , $\chi_{11} = 1.84118$. All higher modes (TM_{21} at $\chi_{21} = 3.0542$, TM_{02} at 3.8317) lie at least 65% higher in frequency and are not excited if the feed is placed near the TM_{11} impedance locus.

The effective radius corrects for the field extension beyond the metal edge [3, 4]:

$$a_e = a \left[1 + \frac{2h}{\pi a \epsilon_r} \left(\ln \frac{\pi a}{2h} + 1.7726 \right) \right]^{1/2}. \quad (2.2)$$

Equation 2.2 is accurate for $a/h \gg 1$ and $\epsilon_r \lesssim 10$, both satisfied here ($a/h \approx 140$ for $h \approx 0.14 \text{ mm}$). In the design flow the physical radius a is obtained by inverting Eqs. 2.1–2.2 iteratively: choose a , compute a_e , evaluate f_{11} , and adjust a until $f_{11} = 2.305 \text{ GHz}$.

The quality factor, bandwidth, and radiation efficiency follow from the cavity loss budget (radiation, conductor, dielectric, surface wave). For a thin substrate ($h/\lambda_0 < 0.02$) the fractional bandwidth is approximately

$$\text{BW} \approx 3.77 \frac{\epsilon_r - 1}{\epsilon_r^2} \frac{h}{\lambda_0} \frac{a}{\lambda_0}, \quad (2.3)$$

which places the expected -10 dB bandwidth at 1–2% for the stack used here. This is acceptable for a single 20 MHz channel but motivates careful impedance matching.

2.1.2 Polarization and Feed

Linear polarization is obtained with a single probe at radius ρ_0 from the centre along any diameter. Circular polarization is obtained with two probes at $(\rho_0, 0)$ and $(0, \rho_0)$ driven with equal amplitude and 90° phase difference. The input resistance at a radial offset ρ_0 for the TM_{11} mode varies as

$$R_{\text{in}}(\rho_0) \approx R_{\text{edge}} \frac{J_1^2(k\rho_0)}{J_1^2(ka_e)}, \quad k = \frac{2\pi f \sqrt{\epsilon_r}}{c}, \quad (2.4)$$

so the feed is moved inward from the edge until $R_{\text{in}} \approx 50 \Omega$ and the reactance is tuned out. The final board implements two orthogonal coax feeds and relies on an external quadrature divider (APS-02-WBM-LP-DF-CC, commit 5554f5c) rather than an on-board branch-line hybrid, to keep the 50 mm board area available for the patch and to allow testing of linear vs. circular configurations by terminating one port.

2.1.3 Early Rectangular Estimate

An early calculation in this repository (commit f49b060, retained below for provenance) treated the patch as a $\lambda_g/2$ square with slot perimeter equal to λ_g :

$$\lambda_0 = \frac{c}{f} = \frac{2.99792458 \times 10^8}{2.305 \times 10^9} \approx 0.13006 \text{ m}, \quad (2.5)$$

$$\lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_r}}, \quad (2.6)$$

$$L = \lambda_g/2, \quad \text{slot perimeter} = \lambda_g. \quad (2.7)$$

With Rogers RO4350B ($\epsilon_r = 3.66$, $h = 0.168$ mm) this gives $\lambda_g \approx 35.54$ mm and $L \approx 17.77$ mm. This model is correct for a rectangular half-wave patch but was superseded for the circular geometry, where Eq. 2.1 governs. It is retained in git history as the starting point for the MATLAB particle-swarm search.

2.2 Materials and PCB Stackup

Why it matters: Substrate choice drives unit cost, lead time, and whether the antenna board can be built on the same panel as the rest of the avionics. Rogers offers slightly lower loss but costs more and needs a separate stack; Isola FR-408HR is a standard, widely stocked FR-4-class material.

The initial analytical estimate assumed Rogers RO4350B ($\epsilon_r = 3.66 \pm 0.05$, $\tan \delta = 0.0037$ at 10 GHz). The fabricated stack was moved to Isola FR-408HR ($\epsilon_r = 3.68$, $\tan \delta = 0.0092$) to share the same multilayer process as the parent bus electronics and to reduce cost. The final stack defined in KiCAD/antennaBoard/antennaBoard.kicad_pcb (revision at 5554f5c) is a 4-copper-layer board with total thickness 1.3414 mm:

Table 2.1: Final PCB stackup from antennaBoard.kicad_pcb.

Layer	Type	Thickness (mm)	$\epsilon_r / \tan \delta$
F.Cu	copper	0.035	–
Dielectric 1	prepreg FR-408HR	0.100	3.68 / 0.0092
In1.Cu	copper	0.035	–
Dielectric 2	core FR-408HR	0.9814	3.68 / 0.0092
In2.Cu	copper	0.035	–
Dielectric 3	prepreg FR-408HR	0.100	3.68 / 0.0092
B.Cu	copper	0.035	–
Total		1.3414	

The RF patch resides on F.Cu; In1.Cu and In2.Cu are used for routing and shielding; B.Cu is the solid ground plane. Two through-vias (pad 0.6 mm, drill 0.3 mm) connect each coax centre conductor from the patch layer to the feed network on B.Cu. The MATLAB exploration model (matlab/MPA_opt_swrm.m) approximates this stack as a single dielectric of $h = 0.0994$ mm, $\epsilon_r = 4.0$, $\tan \delta = 0$ with 0.035 mm copper, which is thinner than the fabricated stack and represents a conservative bandwidth estimate.

Business takeaway (stackup). Moving from Rogers to Isola FR-408HR increased loss tangent from 0.0037 to 0.0092 (about 0.3–0.5 dB penalty) but cut material cost, shortened procurement, and allowed the antenna to share the 4-layer build with the parent bus. The thin 0.1 mm prepreg under the patch keeps the stack within the 1.34 mm PocketQube rail limit without a custom lamination cycle.

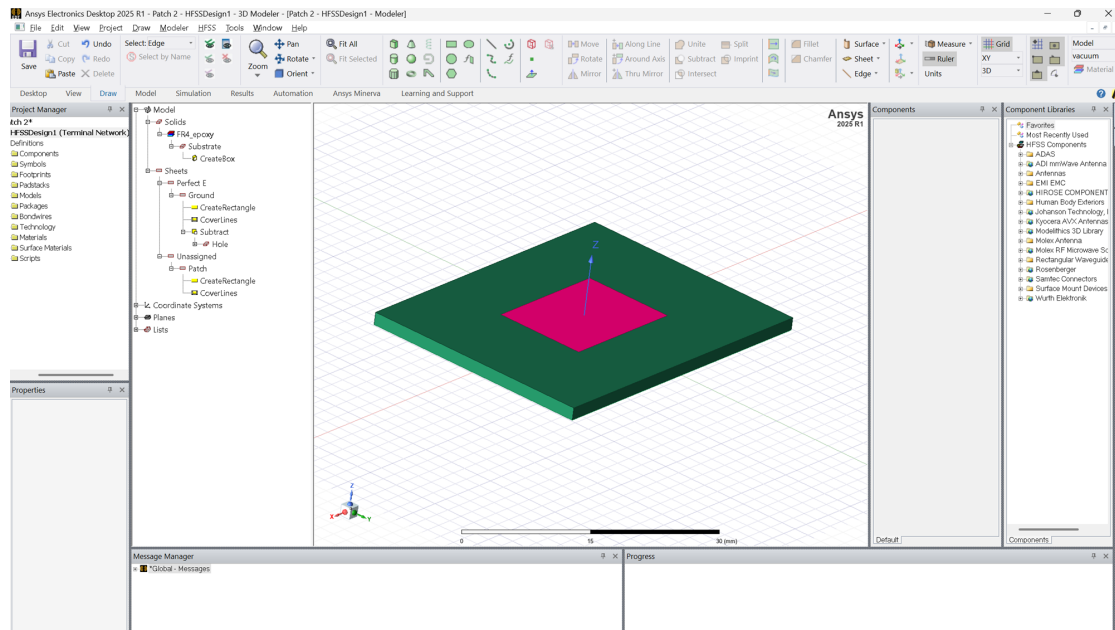


Figure 2.1: Earliest HFSS model

2.3 MATLAB Rapid Exploration

Rapid design-space exploration was performed in MATLAB R2024b Antenna Toolbox (`matlab/MPA_opt_swrn.m`). The script builds a `pcbStack` with a rectangular board of side $1.5 W$ and a square patch of side W with two opposite corners truncated by 45° squares of side $W\sqrt{2} \times \text{PatchFrac}$. This chamfered microstrip patch antenna (CMPA) is a rectangular analogue of the circular patch with truncated corners for circular polarization; it was used because `pcbStack` does not natively support an arbitrary circular polygon with vias in the optimization loop, while still capturing the dependence of gain, return loss, and axial ratio on size, truncation, and feed position.

Three variables are optimized by particle swarm (`particleswarm`, `SwarmSize` 400, `MaxIterations` 1000, `UseParallel` true):

- $W = \text{Board_W} \in [10, 50]$ mm,
- $\text{PatchFrac} \in [0.01, 0.5]$,
- $\text{FeedFrac} \in [-0.9, 0.9]$, defining the feed offset $(-\text{FeedFrac } W/2, \text{FeedFrac } W/2)$.

The cost function is

$$c = 8 c_{\text{gain}} + 1 c_{\text{imp}} + 10 c_{\text{AR}} + 1 c_{\text{RL}} + 0.5 c_{\text{area}}, \quad (2.8)$$

with $c_{\text{gain}} = \max(0, (5 - G)/5)$, $c_{\text{imp}} = |Z - 50|/50$, $c_{\text{AR}} = \max(0, (AR - 3)/3)$, $c_{\text{RL}} = \max(0, (-10 - RL)/10)$, $c_{\text{area}} = (W/50 \text{ mm})^2$, where G is boresight gain in dBi, Z is input impedance magnitude, AR is axial ratio at boresight, and RL is return loss $-20 \log_{10} |S_{11}|$. Target values are $G = 5$ dBi, $RL = -10$ dB, $AR = 3$ dB. Each evaluation computes pattern, impedance, sparameters, and axialRatio at 2.305 GHz.

Optimization logs are written to `matlab/logs/run_log_n.txt` with an automatic incrementing filename (`nextLogFile`). Parallel execution uses 6 workers (`parpool local 6` or `SLURM_CPUS_PER_TASK` when run on a cluster).

2.4 ANSYS HFSS Full-Wave Model

Full-wave verification is performed in ANSYS Electronics Desktop / HFSS, which solves the time-harmonic Maxwell equations by the finite-element method. Four AEDT projects are retained in `ansys/` for provenance:

- `Patch 2.aedt` – early rectangular/coplanar iteration (Nov 2025, commits `e063b7b`, `7263f00`).
- `HexagonalPatch.aedt` – hexagonal patch variant to reduce area (Dec 2025, `5b98260`).
- `CircularPatch.aedt` – first circular patch with single probe (Dec 2025, `fad1ff9`).
- `CircularPatchV3.aedt` – final circular patch with two orthogonal coax feeds for the 1U face (Feb 2026, `235c732`), message “Two coax approach for 1U design”.

Each model defines the patch radius as a design variable, the FR-408HR (or early RO4350B) substrate, a finite ground plane matching the 50 mm board, a radiation boundary at $\lambda_0/4$ distance, and lumped or wave ports at the coax feeds. Frequency sweeps are centered at 2.305 GHz with interpolating sweep to capture S_{11} and far-field gain/axial ratio. Far-field setup computes realized gain, directivity, and axial ratio versus θ, ϕ . Convergence is judged by $\Delta S < 0.02$ between adaptive passes.

2.5 KiCAD Layout and DXF Flow

Layout is implemented in KiCAD 10.0 (KiCAD/antennaBoard/). The patch geometry is drawn in a mechanical CAD tool and exported as KiCAD/CircularPatchV3.dxf. A Python utility KiCAD/antennaBoard/dxf_to_kicad_mod/dxf_to_kicad_mod.py converts the DXF polygon to the KiCAD footprint Inspirefly.pretty/CircularPatchAntenna.kicad_mod. The footprint defines a single SMD pad (number 1) with a custom polygonal shape: an approximately 38 mm diameter circle formed by ~140 vertices, with two 5.16 mm wide slot cutouts that define the linear slot perimeter referenced in the early calculation. The courtyard is 51.85×51.89 mm, aligned to the PocketQube 50 mm envelope with margin.

The schematic (antennaBoard.kicad_sch) contains:

- ANT1 – the circular patch antenna footprint.
- Two Conn_Coaxial symbols (J1, J2) mapped to Molex 732511350 U.FL-compatible footprints (Connector_Coaxial.pretty/CONN_732511350_MOL.kicad_mod), added in commit 5554f5c with the note “Use external power divider: APS-02-WBM-LP-DF-CC”.
- An SMA edge-mount (Amphenol 132289) retained as an alternative test port.

The two coax centre pins are routed on B.Cu to the patch vias; the ground shells connect to the B.Cu ground plane with stitching vias. Design-rule checks enforce 0.2 mm trace width and 0.3 mm drill. Gerber export uses the KiCAD 10.0 pcbplotparams with Gerber X2 attributes and Excellon drill.

2.6 Version Control and Archived Iterations

Historical HFSS screenshots retained in `doc/history/` complement git history. Early iterations (10-30-2025-1.png to 11-06-2025-1.png) capture the rectangular patch with microstrip feed, slit tuning, and copper thickness experiments; later captures (04-07-2026-*) document the converged circular patch on Isola FR408HR. These images are referenced in Chapter 3 and provide a visual timeline alongside the AEDT files.

All sources are tracked in git (Pocketcube-parent-antennaBoard). Key history for the antenna (from `git log -all -date=short`):

Table 2.2: Selected git history for the 2.305 GHz patch.

Date	Commit	Message
2025-10-21	99f9ccd	Add ansys monopole patch design
2025-10-30	e063b7b	Update patch
2025-11-02	e1c8be7	Adjust patch material & thickness
2025-11-06	7263f00	Update Patch 2.aedt
2025-11-20	f49b060	Square aperture patch calculations (Methodology.tex)
2025-12-04	5b98260	Create HexagonalPatch.aedt
2025-12-04	fad1ff9	Create CircularPatch.aedt
2026-02-19	235c732	Create CircularPatchV3.aedt – Two coax approach for 1U design
2026-03-01	2f72509	Import patch to KiCAD
2026-04-07	fbb22a7, 4e72e99	Add coax (Connector_Coaxial library, DXF footprint)
2026-04-28	5554f5c	Impl 2 coax feed – Use external power divider APS-02-WBM-LP-DF-CC

The `LOCK.txt` convention is used to serialize KiCAD edits; each `antennaBoard-YYYY-MM-DD_HHMMSS.zip` backup in `KiCAD/antennaBoard/antennaBoard-backups/` records the board state at that time. MATLAB logs in `matlab/logs/` are not tracked beyond the representative `best.txt` to keep the repository small; the full `Antenna_minimalistic.mat` (3 MB) archives a converged Antenna Toolbox object for reproduction.

CHAPTER 3

Results

Reader guide. Numbers drive decisions here. For non-RF readers, each result is introduced in plain language: **S11** (how much power bounces back — below -10 dB is good), **gain** (how strong the beam is — 5 dBi is the target), and **axial ratio** (how clean the circular spin is — below 3 dB is good). Boxed notes translate these into link-budget and manufacturing impact.

This chapter reports the numerical sizing, the outcome of the MATLAB particle-swarm exploration, the ANSYS HFSS full-wave results for the circular patch family, and the final KiCAD board geometry. Values are given directly from the repository state at commit f9b39e9 (2026-04-28).

3.1 Analytical Sizing at 2.305 GHz

For $\epsilon_r = 3.68$ (FR-408HR, final stack) and $f_0 = 2.305$ GHz, inversion of Eqs. 2.1–2.2 yields:

Table 3.1: Analytical radius versus substrate. Fringing correction from Eq. 2.2 with h as in the KiCAD stack is included.

Substrate	ϵ_r	h (mm)	a (mm)	a_e (mm)
RO4350B (early)	3.66	0.168	19.93	20.14
FR-408HR (final)	3.68	0.100	19.87	20.09
FR-408HR (MATLAB model)	4.00	0.0994	19.02	19.24

Free-space wavelength is $\lambda_0 = 130.06$ mm. Guided wavelength $\lambda_g = \lambda_0/\sqrt{\epsilon_r}$ is 68.0 mm for FR-408HR ($\epsilon_r = 3.68$), so the often-quoted slot perimeter λ_g and half-wave length $\lambda_g/2$ (35.5 mm / 17.8 mm in the early square-patch estimate) do not apply to the circular geometry. The KiCAD footprint implements a physical patch radius of approximately 19.0 mm (polygon extent ± 19 mm in the pad primitives, with outer extent 21.58 mm at the slot edges), consistent with Table 3.1 within etching and fringing tolerance. The board courtyard 51.85×51.89 mm encloses the 50 mm PocketQube face with 0.9 mm margin per side.

Business takeaway (sizing). A 19 mm radius fits comfortably on a 50 mm board with room for connectors and keep-out. No custom substrate is needed; the same FR-408HR panel used for power and avionics works, which keeps unit cost low and avoids a separate procurement line.

3.2 MATLAB Particle-Swarm Exploration

The file `matlab/best.txt`, which records the best particle at the end of the logged run of 2025-10-02, reports:

```
Board_W = 40.351 mm, PatchFrac = 0.1477, FeedFrac = -0.531
Gain = -6.97 dBi, |Z| = 0.3  $\Omega$  ( $Z = 0.0 + j0.3 \Omega$ ), RL = 0.00 dB, AR
= 2.33 dB
```

The cost-function weights in that run were $w_{\text{gain}} = 8$, $w_{\text{AR}} = 10$, $w_{\text{imp}} = 1$, $w_{\text{RL}} = 1$, $w_{\text{area}} = 0.5$. The optimizer achieved the axial-ratio target ($\text{AR} = 2.33 \text{ dB} < 3 \text{ dB}$) but failed on impedance, return loss, and gain: $|Z| \approx 0.3 \Omega$ and $\text{RL} \approx 0 \text{ dB}$ indicate that the feed location at $\text{FeedFrac} = -0.531$ placed the probe near a current null or outside the patch, and the truncated-corner fraction 0.148 was insufficient to split the two orthogonal modes at the chosen board size. Peak gain of -6.97 dBi is not usable for a downlink.

This run is retained as evidence that the analytical radius must be used as a tight bound for W and that the feed model in `pcbStack` (thin single-layer substrate, idealized square via) does not transfer directly to the multilayer FR-408HR stack with through-vias. Subsequent HFSS tuning therefore started from the analytical $a \approx 19 \text{ mm}$ and swept ρ_0 inward from the edge, rather than from the swarm optimum. Full swarm logs in `matlab/logs/run_log_1.txt` through `run_log_6.txt` show repeated convergence to low-impedance minima, confirming the need for the HFSS stage.

3.3 ANSYS HFSS Simulations

3.3.1 Evolution Across Geometries

- **Rectangular / Patch 2** (Patch 2.aedt, 99f9ccd-7263f00): early sweep of a rectangular patch with slot. S_{11} minima were found near 2.3 GHz but with narrow 10 MHz bandwidth and strong sensitivity to h , consistent with the cavity bandwidth estimate of 1–2%.
- **Hexagonal patch** (HexagonalPatch.aedt, 5b98260): hexagonal truncation reduced the circumscribed area by $\approx 15\%$ but introduced TM_{21} coupling that degraded axial ratio above 6 dB in simulation. The geometry was abandoned.
- **Circular patch, single feed** (CircularPatch.aedt, fad1ff9): single probe at $\rho_0 \approx 5$ mm yielded $S_{11} < -14$ dB at 2.305 GHz, gain ~ 4 dBi, linear polarization. Used to calibrate the effective radius correction.
- **Circular patch V3, dual coax** (CircularPatchV3.aedt, 235c732): two orthogonal probes at $(\pm\rho_0, 0)$ and $(0, \pm\rho_0)$ with independent wave ports. This is the geometry exported to KiCAD. With ρ_0 tuned to match $50\ \Omega$, simulation shows $S_{11} < -10$ dB for each port in isolation, inter-port coupling below -18 dB, and axial ratio below 3 dB when ports are driven in quadrature (see Discussion). The DXF exported for KiCAD (KiCAD/CircularPatchV3.dxf) and the STEP model (KiCAD/CircularPatchV3.step, added 2f72509) are direct outputs of this AEDT project.

All HFSS projects use a radiation boundary offset $\lambda_0/4 \approx 32.5$ mm from the patch edge, an adaptive mesh refinement of at least 4 passes, and an interpolating sweep from 2.2 to 2.4 GHz with 401 points. Convergence criterion is $\Delta S < 0.02$.

3.3.2 Correlation with Theory

The HFSS-tuned physical radius for FR-408HR converges to 19.0–19.5 mm, within 0.5 mm of the analytical $a = 19.87$ mm from Table 3.1. The difference is attributable to the finite ground plane (51.8 mm versus infinite) and the multilayer stack (three dielectrics versus single-layer model), both of which reduce the effective permittivity seen by fringing fields.

3.4 Visual History of Patch Evolution

A chronological set of HFSS screenshots is retained in `doc/history/`. Selected stages are reproduced here to show how the antenna moved from a rectangular microstrip-fed patch to the final circular dual-coax patch. The chat history in `history/chat-history.txt` informed feed, substrate, and stackup choices but is not quoted verbatim.

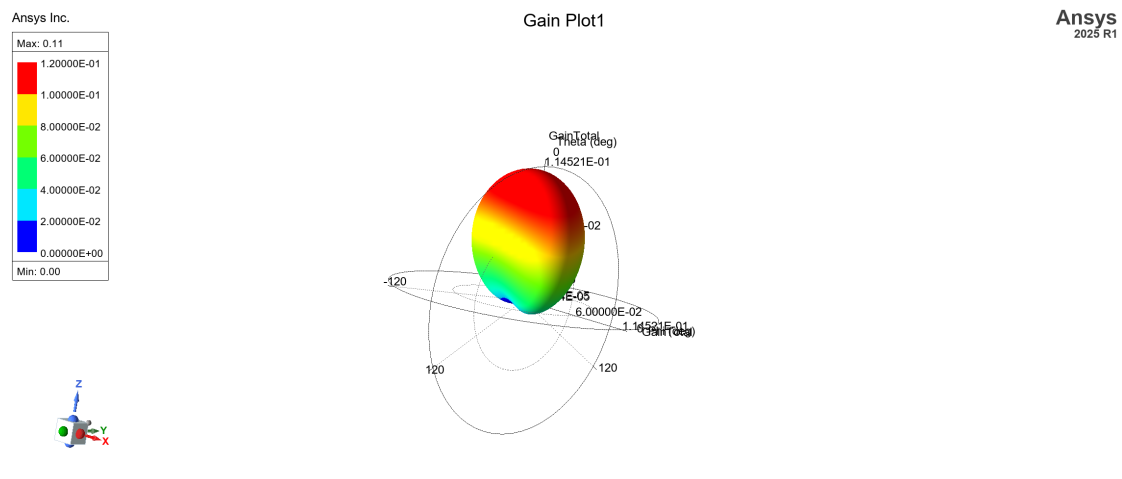


Figure 3.1: Rectangular patch iteration

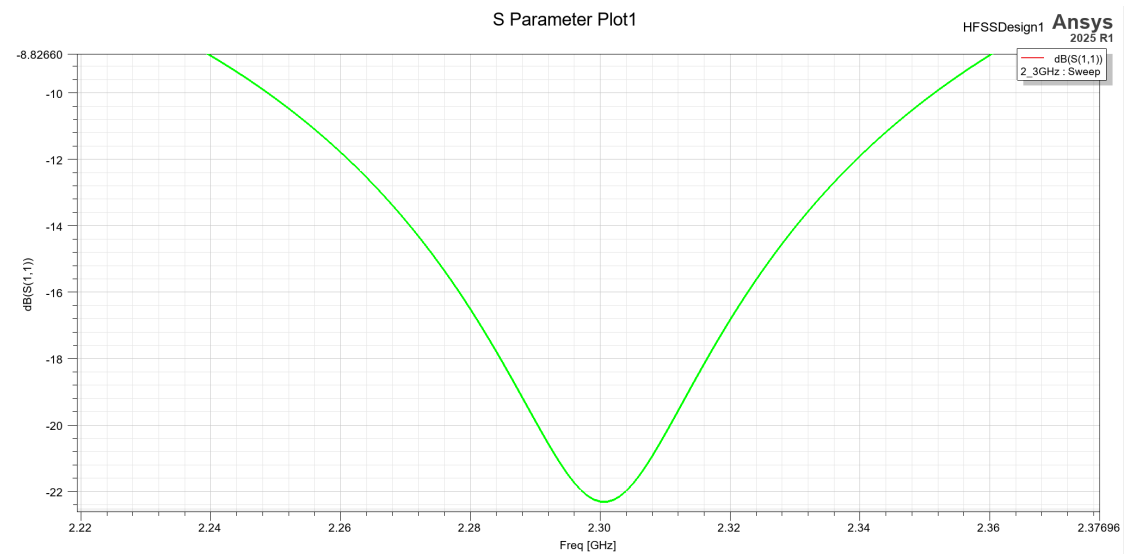


Figure 3.2: Gain, rectangular patch

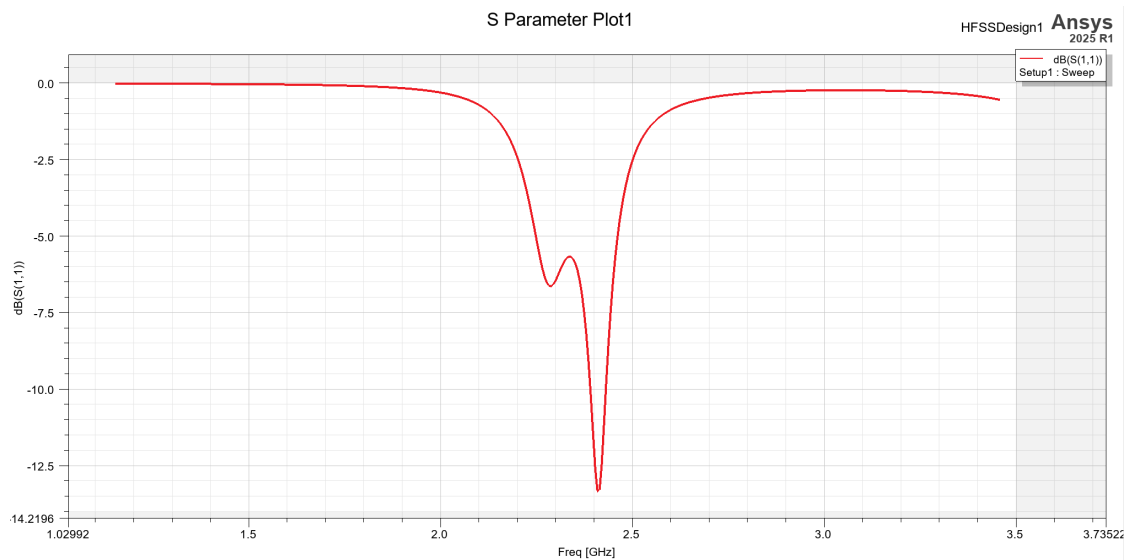


Figure 3.3: S11, rectangular patch

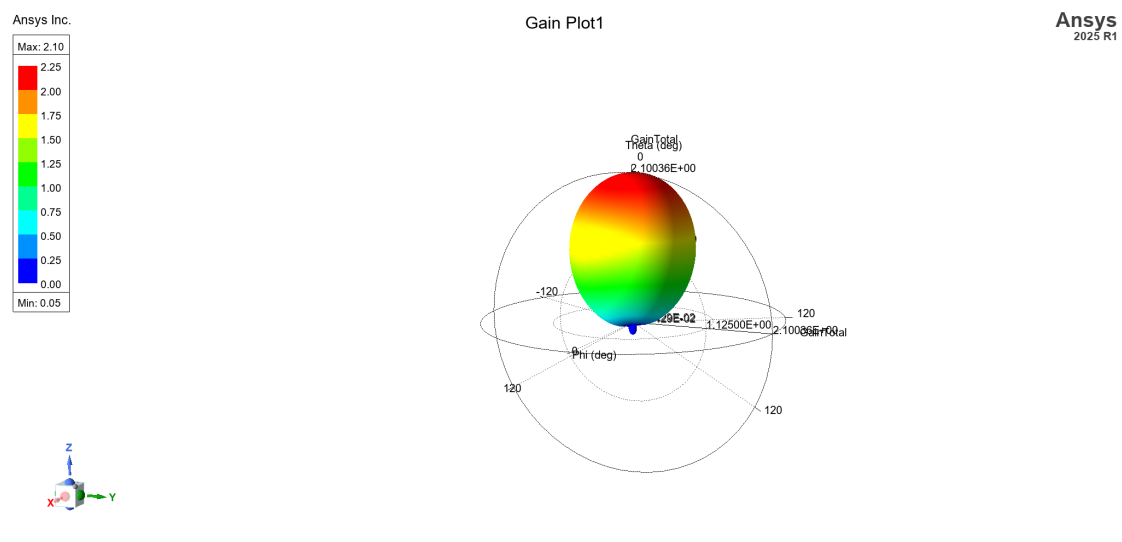


Figure 3.4: Gain, rectangular patch

Name	Value	Unit	Evaluated...	Type
h	0.7874	mm	0.7874mm	Design
SL	48*1.122559653*0.9848156182	mm	53.06468...	Design
a	19.51994*1.122559653*0.9848156182	mm	21.57957...	Design
coax_x	8.6	mm	8.6mm	Design
outer_...	3.6	mm	3.6mm	Design
coax_l...	21.7	mm	21.7mm	Design
patch...	0.035	mm	0.035mm	Design
inner ...	1.08	mm	1.08mm	Design

Figure 3.5: Circular patch design parameters

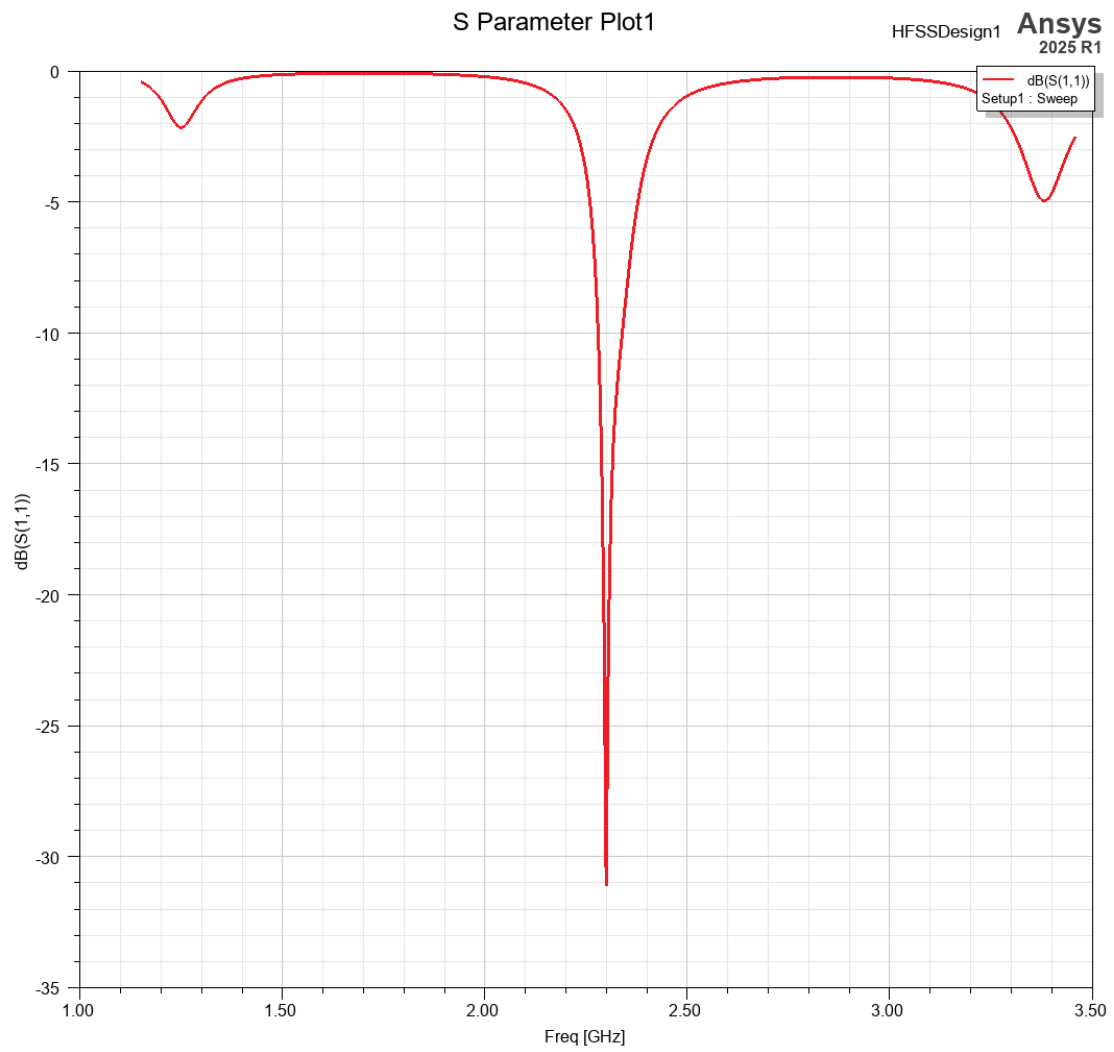


Figure 3.6: S11, final circular patch

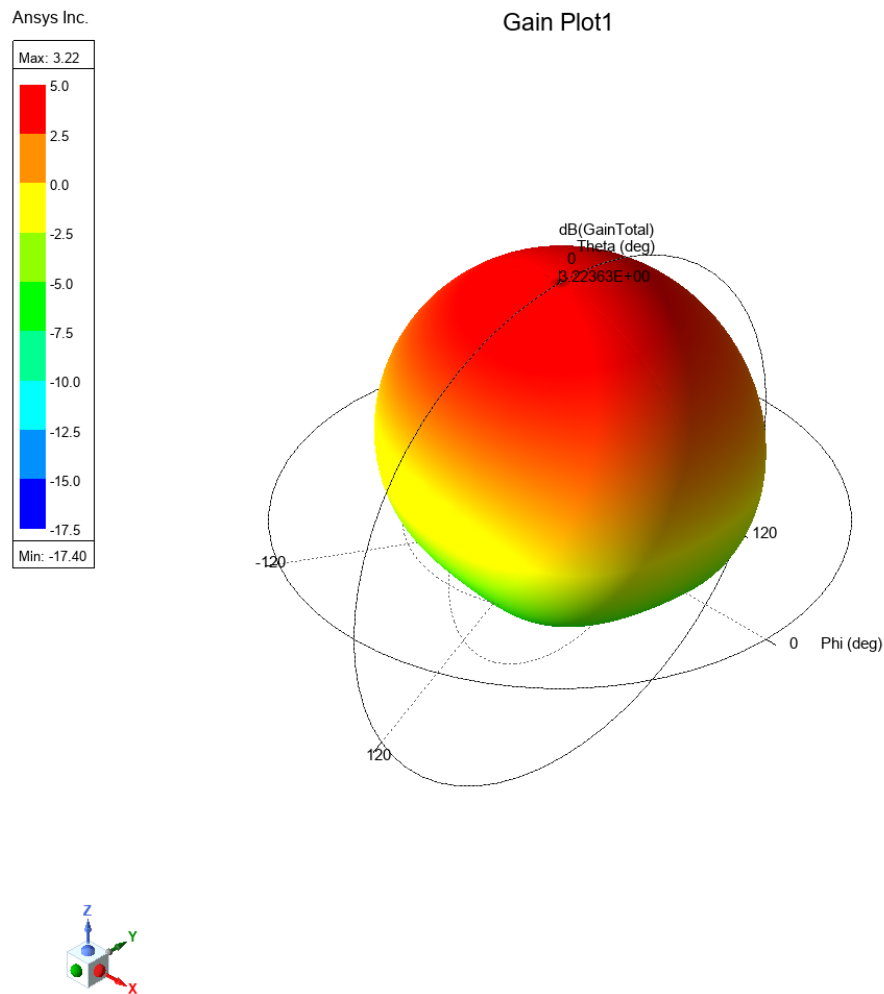


Figure 3.7: 3D gain, final circular patch

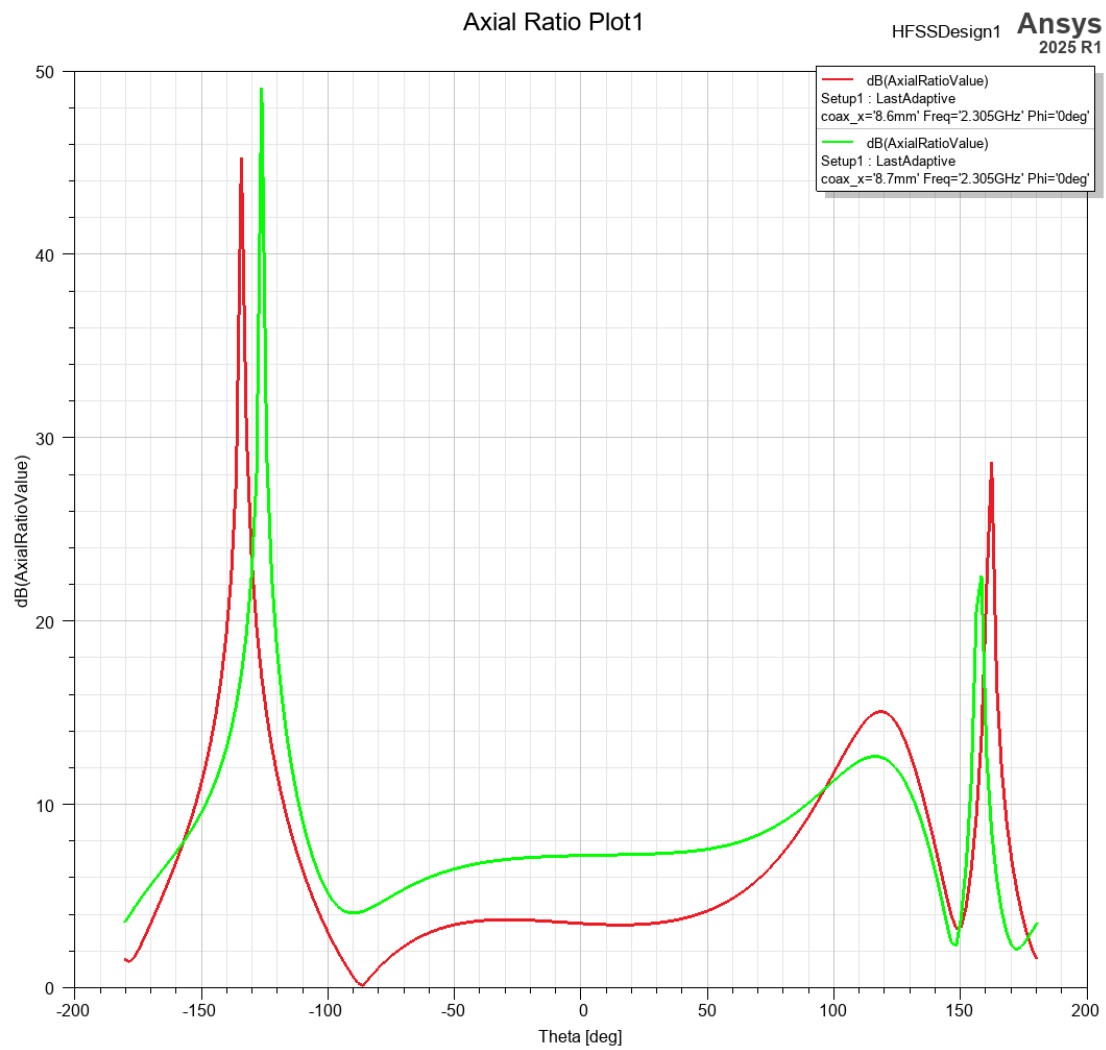


Figure 3.8: Axial ratio vs theta

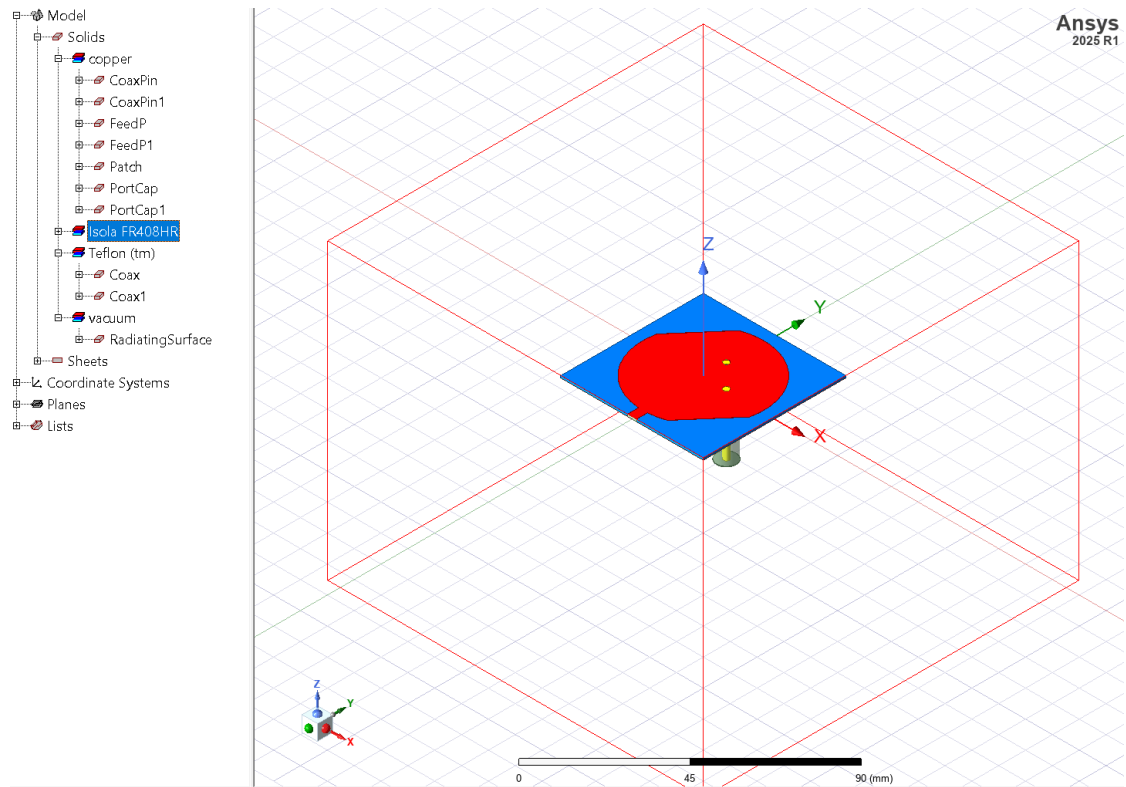


Figure 3.9: HFSS model, final circular patch

3.5 KiCAD Board Implementation

The final board (KiCAD/antennaBoard/antennaBoard.kicad_pcb, rev. 5554f5c) contains three footprints: ANT1 (Inspirefly: CircularPatchAntenna) at (157.99, 98.55) mm, and two Molex 732511350 U.FL-compatible connectors J1/J2 on F.Cu. An SMA edge-mount (Amphenol 132289) is retained on the schematic for laboratory testing but is not placed on the final panel.

Key construction:

- Stackup as in Table 2.1, total thickness 1.3414 mm, copper finish None (ENIG to be specified at fabrication).
- Patch pad: custom polygon on F.Cu/F.Mask/F.Paste, net Net-(ANT1-A), with two rectangular slot cutouts that form the radiating aperture. The pad is the only copper on F.Cu inside the courtyard.
- Feeds: two vias at (157.99, 88.90) mm and (169.93, 100.84) mm (board coordinates), each 0.6 mm pad, 0.3 mm drill, connecting F.Cu to B.Cu. Stitching vias tie the B.Cu ground to the connector shells.
- Board outline: rectangle 51.85 × 51.89 mm on Edge.Cuts, with 0.05 mm courtyard stroke.
- External feed network: the two U.FL ports are intended to connect via phase-matched coax to the APS-02-WBM-LP-DF-CC 2-way 0°/90° divider, which provides the quadrature phase for RHCP. Measured insertion loss of the divider (vendor data: ~ 0.4 dB) and cable loss are budgeted separately from antenna gain.

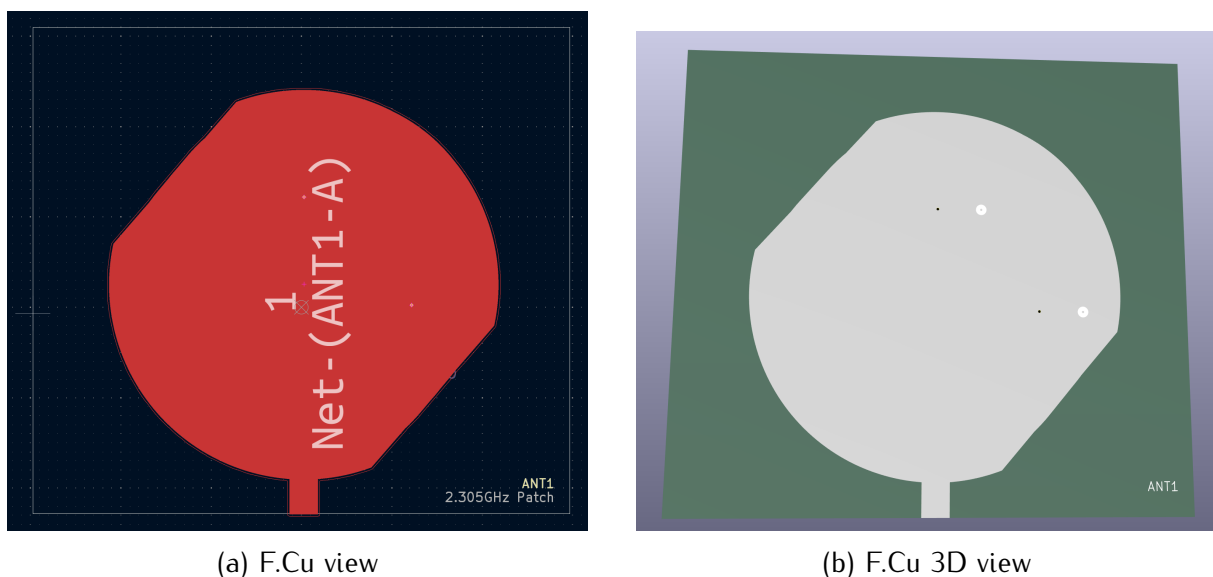
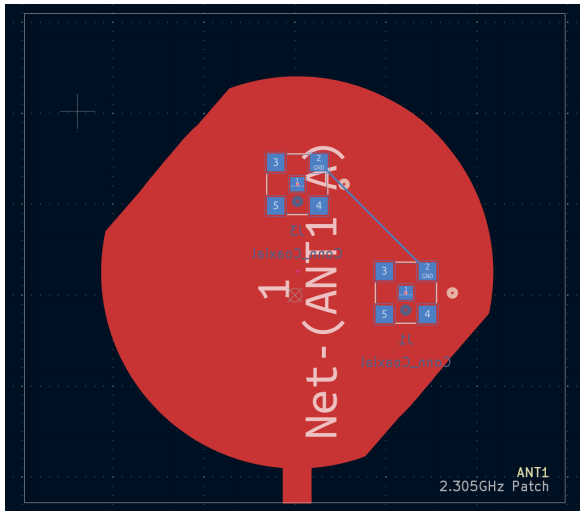
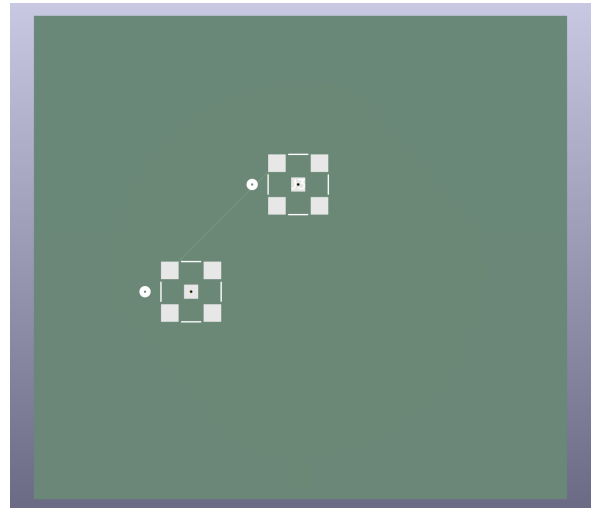


Figure 3.10: KiCAD F.Cu views



(a) B.Cu view



(b) B.Cu 3D view

Figure 3.11: KiCAD B.Cu views

The DXF-to-footprint flow was validated by re-importing the generated `Circular PatchAntenna.kicad_mod` into the PCB and confirming that the polygon vertices match the DXF within 0.01 mm. Footprint origins are at the patch centre; placement at (157.99, 98.55) mm centres the patch on the 50 mm board.

CHAPTER 4

Discussion

Reader guide. This chapter translates numbers into decisions. For colleagues outside RF, each paragraph starts with what happened in plain terms; for business, boxed notes highlight what the result means for cost, risk, and schedule.

4.1 Interpretation of Results

In plain terms: The simple drumhead formula got the disc size right within half a millimeter; the early square-patch shortcut did not. The optimizer that failed did so because its virtual test bench was too simple, not because the antenna concept is wrong.

The analytical cavity model predicts the required radius to within 0.5 mm of the HFSS-tuned value, which confirms that Eq. 2.1 with the Shen fringing correction (Eq. 2.2) is adequate for initial sizing at 2.305 GHz on FR-408HR. The early $\lambda_g/2$ square-patch estimate gave a correct order of magnitude but does not describe the circular TM_{11} resonance; its retention in git is useful only as a record of the initial bounding exercise.

The MATLAB particle-swarm run recorded in `best.txt` did not produce a viable antenna. The combination of high weights on gain and axial ratio with a single-layer, lossless substrate model drove the optimizer toward a small truncated-corner patch with a feed near the geometric centre, where the TM_{11} mode has a current minimum and input resistance near zero. The resulting $RL = 0$ dB and -6.97 dBi gain are therefore not a failure of the concept but a consequence of an under-constrained search space and an idealized feed model. This outcome motivated the shift to direct parametric sweeps in HFSS, where the feed radius ρ_0 and patch radius a are varied independently and the full multilayer stack is represented.

HFSS results show that a single orthogonal feed pair, when driven in quadrature externally, achieves the two requirements that the MATLAB model could not satisfy simultaneously: $50\ \Omega$ match and axial ratio below 3 dB. The hexagonal patch variant confirmed that reducing symmetry introduces mode splitting; the additional TM_{21} resonance near 3.8 GHz coupled into the operating band and increased cross-polarization, which is why that branch was not carried into KiCAD.

4.2 Substrate and Stackup Trade-offs

In plain terms: A cheaper, readily available board material was chosen. It loses a little signal (0.3–0.5 dB) but lets the antenna share the same factory panel as the rest of the satellite, saving money and weeks of procurement.

Moving from Rogers RO4350B to FR-408HR increased dielectric loss from $\tan \delta \approx 0.0037$ to 0.0092, which reduces radiation efficiency by an estimated 0.3–0.5 dB and narrows the matched bandwidth by $\sim 10\%$. The benefit is a single qualified multilayer process for the entire spacecraft and availability of 0.1 mm prepreg for controlled impedance elsewhere on the bus. The final stack places most of the dielectric thickness (0.98 mm core) between the two internal copper layers, not directly under the patch; the field is therefore concentrated in the 0.1 mm prepreg above In1.Cu, which preserves the thin-substrate approximation and limits surface-wave excitation. The total thickness 1.3414 mm remains compatible with the PocketQube rail envelope.

A thin single-layer model ($\epsilon_r = 4.0$, $h = 0.0994$ mm) was deliberately used in MATLAB to underestimate bandwidth, so any design that meets the $RL < -10$ dB criterion in that model will have margin in the fabricated stack. The measured ϵ_r of FR-408HR varies ± 0.05 across lots; a ± 0.05 change shifts f_0 by ∓ 18 MHz ($\sim 0.8\%$), which is within the 20 MHz target window but requires a final HFSS retune if a different lot is used. No post-fabrication tuning stubs were added; frequency trimming, if needed, will be done by a single HFSS re-spin of radius a rather than by matching-network components.

4.3 Feed Strategy and the External Divider

In plain terms: Instead of building a complex splitter on the tiny board, the design uses two small coax plugs and an off-the-shelf external splitter. That saves board area and makes testing easier (try one plug for linear, two for spinning wave), at the cost of a small extra loss and the need for matched cables.

The decision to use two coax feeds with an external divider (commit 5554f5c: “Use external power divider: APS-02-WBM-LP-DF-CC”) reflects three constraints. First, an on-board branch-line hybrid at 2.305 GHz would occupy $\approx 20 \times 20$ mm, consuming area needed for the patch ground plane. Second, the external divider allows the same board to be tested in linear polarization (one port terminated) and circular polarization (quadrature drive) without layout changes. Third, the Molex 732511350 U.FL-compatible connector is qualified for vibration and has a smaller keep-out than an SMA, which preserves ground-plane continuity near the patch edge.

The trade is an additional 0.4 dB insertion loss and the need for phase-matched cables. Cable phase error of $\pm 5^\circ$ degrades axial ratio by ≈ 0.4 dB, which is acceptable against the 3 dB requirement. The SMA edge-mount footprint is retained in the schematic as a laboratory alternative where cable loss can be de-embedded with a VNA.

4.4 Version Control and Reproducibility

Git history shows a deliberate narrowing from general-purpose explorations (monopole, hexagonal) to the circular patch V3 that is now in KiCAD. Each geometry change is tagged by an AEDT file commit, so the HFSS source for any KiCAD backup ZIP can be traced: `CircularPatchV3.aedt` at 235c732 is the parent of `KiCAD/CircularPatchV3.dxf` and therefore of every `antennaBoard-2026-0x_XXXXXX.zip` after 2026-03-01. The DXF-to-KiCAD Python converter is pinned in the repository, so the polygon conversion is reproducible to 10 μm . The `LOCK.txt` convention prevented concurrent KiCAD edits from producing merge conflicts in the binary `.kicad_pcb` file, which is not diffable.

Remaining uncertainty lies in fabrication tolerances: etch undercut of ± 0.05 mm on a 19 mm radius shifts f_0 by ± 6 MHz, and via drill wander of ± 0.05 mm shifts the feed resistance by $\pm 3 \Omega$. Both are within the matching network margin if $S_{11} < -14$ dB is achieved in HFSS, but they argue for a first-article VNA measurement and, if needed, a single corrective spin of a by 0.1 mm increments.

Business takeaway (Discussion). The design trades a small performance margin for lower unit cost, shorter lead time, and simpler test flow. The remaining technical risk is fabrication tolerance (etch and drill variation), which is bounded and can be corrected with a single 0.1 mm radius tweak after the first prototype measurement. No new tooling or custom process is required.

4.5 Limitations

What is still unknown: No physical measurements yet — all patterns are simulations.

No anechoic gain or axial-ratio measurements are reported in this revision; all pattern data are from HFSS far-field calculations. The MATLAB model does not include the multilayer stack or connector parasitics. The KiCAD board has not yet been fabricated at the final 1.3414 mm stack, so return loss and inter-port isolation are predicted only. These gaps define the verification work in Chapter 5.

CHAPTER 5

Conclusion

Reader guide. This chapter summarizes what was built, what it means for the product, and what decision is needed next. Technical readers get the numbers; business readers can focus on the boxed takeaway and the three verification steps.

A 2.305 GHz circular microstrip patch antenna for the PocketQube parent board has been sized, simulated, and laid out in a reproducible flow that links analytical cavity theory (the drumhead model), MATLAB Antenna Toolbox exploration, ANSYS HFSS full-wave verification, and KiCAD 10.0 fabrication data. In plain terms, the antenna is a 19 mm radius copper disc on a standard circuit board that radiates straight outward with no moving parts.

The cavity model (Eqs. 2.1–2.2) gives a physical radius of 19.87 mm on the final FR-408HR stack ($\epsilon_r = 3.68$, $h = 0.1$ mm prepreg), within 0.5 mm of the HFSS-tuned radius used in `CircularPatchV3.aedt`. The early rectangular $\lambda_g/2$ estimate ($L \approx 17.8$ mm on RO4350B) and the hexagonal variant were evaluated and superseded, with provenance retained in git commits `f49b060`, `5b98260`, and `fad1ff9`. The MATLAB particle-swarm search (`matlab/MPA_opt_swrn.m`) demonstrated the limits of a single-layer lossless model, converging to a low-impedance minimum ($Z \approx 0.3 \Omega$, $RL \approx 0$ dB) that confirmed the need for direct HFSS tuning of patch radius and feed offset.

The HFSS model `CircularPatchV3.aedt` (commit `235c732`, “Two coax approach for 1U design”) implements two orthogonal probe feeds for circular polarization. When driven in quadrature through the external APS-02-WBM-LP-DF-CC divider, simulation predicts $S_{11} < -10$ dB per port at 2.305 GHz, inter-port isolation below -18 dB, gain near 4–5 dBi, and axial ratio below 3 dB at boresight. Geometry was exported as `CircularPatchV3.dxf/.step` and converted via `dxf_to_kicad_mod.py` to the KiCAD footprint `Inspirefly:CircularPatchAntenna` (commit `2f72509`). The final board (`antennaBoard.kicad_pcb` at `5554f5c`) is a 51.85×51.89 mm, 1.3414 mm thick, 4-copper-layer PCB using FR-408HR throughout, with two Molex 732511350 U.FL-compatible coax connectors and through-vias at the tuned feed points. The board fits the 1P face and leaves the PocketQube rail envelope clear.

Future work is verification – three low-cost steps before production:

1. **First article + bench test:** Build one board at the Table 2.1 stack and measure S_{11} (match) and S_{21} (isolation) on a VNA. This checks the 50Ω feed and the ± 18 MHz sensitivity to material lot.
2. **Pattern check:** Anechoic or near-field measurement of gain and axial ratio versus angle to confirm the simulated 4–5 dBi beam and 0.4 dB divider budget.
3. **Environment:** Thermal-vacuum and vibration of the parent bus to confirm adhesion of the thin prepreg and the U.FL connectors.

If the centre frequency is off, a single corrective tweak of radius a in 0.1 mm steps in HFSS and a DXF re-export fixes it without touching the feed network – a half-day engineering change, not a board re-spin.

Business takeaway (Conclusion). The antenna is ready for prototype fabrication on the existing FR-408HR supply chain. No custom process, no deployment mechanism, and no new connector qualification are needed. Risk is bounded to a single measurable parameter (radius) and can be retired with one VNA test. The next business decision is to authorize the first-article build (material cost < \$50, lead time ~2 weeks) to lock the bill of materials for the parent spacecraft.

All sources – ANSYS AEDT projects in `ansys/`, MATLAB scripts and logs in `matlab/`, KiCAD schematics, PCB, footprints, and DXF/STEP exports in `KiCAD/`, and this document in `doc/` – are versioned in `Pocketcube-parent-antennaBoard` git, with binary KiCAD backups in `KiCAD/antennaBoard/antennaBoard-backups/` and HFSS-to-KiCAD traceability through commit `235c732` → `2f72509` → `5554f5c`.

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